

USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #5 (Lesson 19) - Beam Steering Lab
Lab Sheet
Fall 2026

Name: _____ Section: _____ Date: _____

Learning Objective 3.5: I can implement beam steering on the ADALM-PHASER and verify the steered pattern against theory.

This sheet is the turn-in document for the Lesson 19 lab. Record the source frequency the GUI reports before you start, fill in every box, and keep the prediction table from Lesson 18 beside you. The lab is a team assignment, so one sheet per team.

- Table A – commanded against physical angle.** Place the source at each position on the arc, hunt for the commanded Steer Angle that maximizes the FFT amplitude, and record it. The last column expresses the difference in sweep grid steps, one step being the ADAR1000's 2.8125° phase LSB.

Physical angle	Commanded angle at peak	Difference	Difference / 2.8125°
0°			
30°			
45°			

- Table B – phase check at 30° .** With the peak command applied, read the eight per-element phases out of Phase Control and compare them against the wrapped ramp predicted in Lesson 18. At $\theta_0 = 30^\circ$ the progressive phase is $\Delta\phi = 176.8^\circ \sin 30^\circ = 88.4^\circ$. All values in the table are in degrees.

Element	1	2	3	4	5	6	7	8
Predicted								
Applied								
Difference								

Half an LSB = Is every difference smaller than that?

- 3. Table C – beamwidth.** Measure the half-power width of the frozen boresight trace and of the 30° trace by reading the angles where each trace falls 3 dB below its own peak, then record the peak-level difference between the two traces.

Quantity	Measured	Calculated
HPBW at $\theta_0 = 0^\circ$		
HPBW at $\theta_0 = 30^\circ$		
Peak level, 30° against 0°		

4. Written answers. Three or four sentences each.

- (a) Why does the commanded angle at which the trace peaks move in steps of about 2.8° rather than continuously, and what in the hardware sets that step size?

- (b) Explain why a plot of received power against commanded steer angle is a picture of the array's pattern at all. Name the principle you are relying on, and say what would change if the source were moving during the sweep.

Documentation: _____